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**Research Article**

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# Local Tool Path Smoothing Based on Symmetrical NURBS Transition Curve with Look Ahead Optimal Method: Experimental and Analytical Study

Chunlin Lei[1], Yusoff Nukman[1], Hwa Jen Yap[1], Ali H. Abdelrazek[2], Mingbing Luo[3]

## Abstract

Linear tool path with the inherent shortcoming of tangential and curvature discontinuities which lead to poor machining efficiency and surface finish. A lot of local corner smoothing methods which inset a tiny spline curve at the corners have been proposed in literature. However, there exists potential overlap of the two adjacent spline curves while smoothing, and still no optimal solution for eliminating the overlaps. To optimally eliminate the overlaps and smooth the transition curve curvature in real-time, a symmetrical Non-uniform rational B-spline (NURBS) curve transition based on look ahead optimal method is proposed in this paper, by scanning a set of unconstrained pre-cornering transition curve information, analysing the overlap type and backward determining the optimal transition length, thus fully use the linear segment length of each corner, Thus fully increasing the feedrate at the corners. $G^2$ continuity, analytical expression of the curvature extreme, approximation error constraint and real-time performance are considered simultaneously in this paper. Simulation and real machining experiments are made to compare with the existing symmetrical rounding algorithms, the results show that the proposed method has a significant curvature decrease of 17.64% and 4.44% improvement of efficiency, and the curvature distribution of the adjacent curves are balanced, thus improved the machining velocity smoothness. In addition, the real machining experiment results validate the proposed method can work well in 125us real-time control period CNC system.

***Keywords:*** *Symmetrical; NURBS; Look ahead; Optimal transition length; Balanced; Real-time*

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**Nomenclature**

| | | | |
|---|---|---|---|
| $L_{left}^{i}$ | The remained length of $i$ th segment after transition | $C_i(u)$ | Transition curve |
| $d_{i,0}$ | Length of control polygon | $P_{i-1},\ P_i,\ P_{i+1},\ P_{i+2}$ | Point of linear tool path |
| $\boldsymbol{a}_{i,0},\ \boldsymbol{a}_{i,1}$ | Unit tangential vector | $Q_{i,0}$ ,$Q_{i,1}$ ,$Q_{i,2}$ ,$Q_{i,3}$ ,$Q_{i,4}$ | Control points of NURBS curve |
| $\beta$ | Corner angle of two adjacent linear segments | $w_{i,0}$ ,$w_{i,1}$ ,$w_{i,2}$ ,$w_{i,3}$ ,$w_{i,4}$ | The weights of control points of NURBS |
| $k_m$ | Curvature extreme of the $i$ th transition curve | $L_{i,0},\ L_{i,1},\ L_{i+1,0}$ | Transition lengths of transition curve |
| $\varepsilon$ | Geometric error of the $i$ th transition curve | $\varepsilon_m$ | The maximum allowed geometric error |

# 1 Introduction

With the rapid development of modern aviation, aerospace, and model industries, a large number of complex curve and surface modelling are used in the parts design. The parametric curves, such as polynomial splines, Bézier, B-spline, and NURBS, are usually used to describe these mechanical parts in computer aided design (CAD) software, which are also the ideal representations of the tool paths in free-form machining. However, most of tool paths are still represented as series of linear segments (G01) and circular segments (G02 or G03) which discretizing the parametric curves under the predefined tolerance in computer aided manufacturing (CAM) software [1]. Due to the tangency and curvature discontinuities of the linear and circular tool paths at the joint points, frequent decelerations and accelerations may lead to poor machining efficiency and accuracy [2]. To solve these problems, smoothing the discontinuous linear segments is very significant for ensuring the high order geometric continuity of the tool paths and then achieving good surface quality together with high processing efficiency [3]. The toolpath smoothing methods can be classified into local smoothing [4, 5] and global smoothing [6, 7] according to the smoothing scope, but in practical, local smoothing is much more stable, the approximation can be control accurately, and real-time implementable than global smoothing.

Typically, the local curve-based methods, which inset a tiny parametric curve to blend the adjacent linear segments, such as Bézier splines [8, 9], B-spline [10, 11], NURBS spline [5, 12], Pythagorean-Hodograph (PH) spline [3, 13], Clothoid curve [14], airthoid spline [15]. According to the shape of the inserted curve, the local smoothing can be classified into symmetrical methods and asymmetrical methods. Most of the symmetrical methods are imposing excessive constraints on the transition lengths or by reducing proportionally their approximation error to the corners with curvature increasing sacrifice, the asymmetrical methods can round the corners with curvature as small as possible within the specified approximation error, but without analytical solution to obtain the curvature extreme make it difficult in real-time implementation.

When smoothing the short-segmented tool paths, to avoid the potential overlaps between adjacent transition curves, the most comment solution is using a fixed-length constraint that the transition length was no more than half the linear segment, which is the so called fixed-length constrained rounding (FLCR) method. To overcome the first-order

discontinuity at the corners of linear segments, with the approximation error constraint, the parameterized synchronization constraint and continuous curvature constraint into consideration, a double $G^2$ continuous cubic Bézier curves was proposed in [9] for five-axis linear tool path, the transition length of both symmetrical curves are limited no more than half the length of the corresponding line. Zhao et al. [16] developed an analytical decoupled corner smoothing method by inserting an asymmetric cubic B-spline and a pair of symmetric quartic spherical Bézier curves for tool tip position and tool orientation respectively. Zhang et al. [5] proposed a double cubic NURBS with 5 control points and related weights, which satisfies the curvature continuous and accuracy constraint for G01 tool path, the control points and weights are parameterized to be symmetrical and overlap was eliminated by mathematically limit the transition length. Zhang et al. [17] desired two quartic B-spline to smooth the tool tip and tool orientation respectively for five-axis linear segments, among them the tool tip position is smoothed with fixed-length constraint symmetrical curve, while the tool orientation is asymmetrical curve. The smoothing error of the tool tip position and tool orientation are mathematically constrained in workpiece coordinate system, $G^3$ continuity is guaranteed along the entire tool path. Considering the kinematics performance of machine tools, a quintic B-spline curve local corner smoothing with the fixed-length constraint was proposed in [10, 18], which can ensure high-order continuity of transition curves and steady motion of machine tools. Huang et al. [15] the first time proposed a “airthoid” curve to smooth the corners of the G01 tool path. The clothoid spline is an arclength-parameterized spline, thus the feedrate fluctuation is eliminated in feedrate scheduling phase, this is an advantage over other none arc-length parameterized spline, but the maximum transition length is also restricted to less than half length of the original linear segment. Yang et al. [19] extended the fix-length constraint quintic B-spline to tool path smoothing of robotic machining and achieved the improvement of robot motion smoothness and tracking accuracy. To archive smaller curvature, different from the traditional corner rounding methods whose control points of transition curves lie in linear segments, Xu and Sun [20] proposed a circumscribed corner rounding method with a part of control points outside of the corner to modify the path curvature and feedrate at the corners. Indeed, simply constraining the transition length can avoid the overlaps between adjacent transition curves, but this fixed-length constraint is excessively restrictive. Under this constraint, even if the transition length exceeds half the linear tool path segment without overlaps, it also has to shrink to the fixed length. For the purpose of alleviating the excessive constraint on transition length, Du et al. [21] a quadratic equation with respect to transition length is reformulated to optimize the problem, the joint-point of two adjacent transition curve is obtained analytically.

Different from the methods that use the fixed-length constraints to solve the overlap problem, some attempts focus on eliminating the overlap by proportionally reducing the approximation error to the corners, which is also known as the proportional adjustment (PA) strategy. Zhao et al. [22] proposed a curvature-continuous B-spline with five control points to blend the short linear tool path .By recursive half one subdivision to proportionally shorten the two adjacent transition length until the overlap eliminated, thus to obtain an suboptimal solution of transition length. To archive higher curvature continuity, Fan et al. [23] proposed a symmetrical $G^3$ continuously quartic Bézier curve with 9 control points to round the corners, the transition curves overlap were eliminate with PA strategy but different from Zhao et al. [22], a small then 1 but lager than 0 factor was used to recursively shorten the two adjacent transition length in same proportion other than half one subdivision, which can get more accurate result if the factor is near to 1, but in sacrifice of computing time.

These methods only consider the adjacent two transition curve, however, the transition length adjusted proportionally also couples possibly with the next adjacent transition lengths when the overlap continuously appears in the tool path. Zhang et al. [24] proposed a method to search the suboptimal scaling factor of each corner, which relates to the transition length and smoothing geometric error of the transition curve, by recursively increase the scaling factor of each transition curve with a small increment until the specified constrains not be satisfied anymore, then frozen this factor, until all the factors respected to each corner have been frozen. This is a good solution for short linear segments; however, it is time consuming especially when dealing with long segments and large smoothing error allowance, and the algorithm accuracy is influenced by the increment.

Recently, different from symmetrical methods that the transition length of both side of the corner must keep the same length, asymmetrical transition methods which can adjust independently the transition length on both sides of corner. Shi et al. [25] proposed an asymmetrical quintic PH spline to smooth the G01 blocks for improving the feedrate fluctuation and machining efficiency, although the transitional lengths at the corners can be obtained analytically with the approximation error, but the potential overlap was not considered especially when dealing with the short-segmented tool path. Wan et al. [3] constructed an asymmetrical PH spline by introducing two steps to shorten the spline and connect the adjacent two splines. In step 1 by shorting the spline whose control polygon shares the longer segment to satisfy a specified equation, if the overlap still occurs after step 1; then in step 2 a proportional factor is implicated to shorten the splines with same proportion thus eliminate the overlap, this step, shorten the adjacent transition length also can be called PA strategy. Yan et al. [11] proposed an unconstrained pre-cornering transition model to analyse the effect of tool path geometry on transition overlap, and classified the overlap into 3 types according to the transition length and the their position in the segments, then two overlap eliminating strategies were adapted to adjust the control points, thus the asymmetrical B-spline was constructed. The approximation error was proved within the user-specified approximation error, but the curvature extreme point of the asymmetrical curve was not given, this is also an inherent short come of asymmetrical curve, there no analytical solution to get the curvature extreme.

Compared to asymmetrical curve, the curvature extreme of symmetrical curve is just in the middle of the curve, it can be analytically expressed; another advantage is to estimate the spline length. An obvious difficulty of smoothing methods based on Bézier, B-spline, or NURBS is that there is no analytical relation between the arc length and its parameter, an iterative numerical algorithm is needed to estimate the curve length before velocity scheduling [26], then for symmetrical curve, the iterative algorithm only need run half of the curve, by doubling the result can get the estimate length of the curve, this is an significant advantage in real-time smoothing and interpretation. However, there are still works to do in symmetrical curve transition, achieving optimal transition length solution in real time. The main contribution of this paper is to design a real time local transition algorithm optimally for CNC machining of long and short segmented tool path. The proposed method can obtain better feedrate profile at the corners by look ahead scanning and backward optimizing, fully use the linear segment length of each corner and avoid curvature extreme polarization at the same time, thus decrease average curvature as much as possible under the specified approximation error constraint, thereby fully increasing the feedrate at the corners. There is no complex mathematical operation in look ahead scanning and backward optimization process, makes it easy to implement in real-time CNC system.

# 2 Symmetrical NURBS curve transition

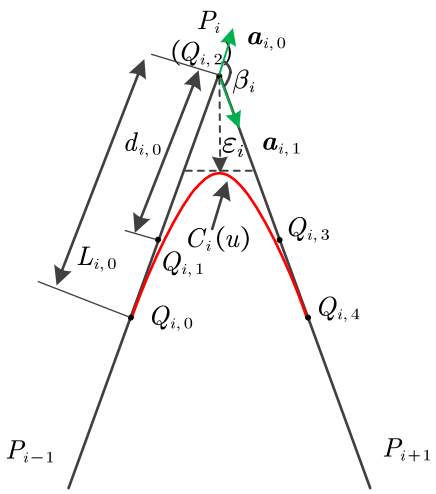


**Fig. 1** The cubic NURBS transition model.

In consideration of $G^2$ continuity which can guarantee the continuities of tangency and curvature so that the fluctuation in feed, acceleration and jerk can be improved greatly [5], a cubic NURBS is adapted in this paper to smooth the linear segment tool path, as illustrate in Fig. 1.

## 2.1 Construction of cubic NURBS curve

Accoding to Les and Wayne [28] the cubic NURBS with five control points in this paper is given as:

$$C_i(u) = \frac{\sum_{j=0}^{4} N_{j,3}(u) w_{i,j} Q_{i,j}}{\sum_{j=0}^{4} N_{j,3}(u) w_{i,j}} \tag{1}$$

$$\begin{cases} N_{j,3}(u) = \dfrac{u - u_j}{u_{j+3} - u_j} N_{j,2} + \dfrac{u_{j+4} - u}{u_{j+4} - u_{j+1}} N_{j,2} \\ N_{j,0}(u) = \begin{cases} 1, \text{ if } u_j \leqslant u \leqslant u_{j+1} \\ 0, \text{ otherwise} \end{cases} \\ \text{define } \dfrac{0}{0} = 0 \end{cases} \tag{2}$$

Where $j = 0, 1, 2, 3, 4$; $Q_{i,j}$ are the control points, $w_{i,j}$ are the weights, particularly, $w_{i,0} = w_{i,4}$, $w_{i,1} = w_{i,3}$, and the node vector is $U = \{0, 0, 0, 0, 0.5, 1, 1, 1, 1\}$. Meanwhile, the control point $Q_{i,2}$ is located on tool point $P_i$, $Q_{i,0}$, $Q_{i,1}$ and $Q_{i,4}$, $Q_{i,3}$ are symmetrically located in linear segments $P_{i-1}P_i$ and $P_iP_{i+1}$, and satisfy the following conditions:

$$L_{i,0} = |Q_{i,0}P_i| = |P_iQ_{i,4}| \tag{3}$$

$$d_{i,0} = |Q_{i,1}P_i| = |P_iQ_{i,3}| \tag{4}$$

where $L_{i,0}$ presents the transition length of the NURBS curve. According to [29] $1.4 \leq L_{i,0}/d_{i,0} \leq 1.7$ gives satisfying result in terms machining time minimization. In this paper the radio was set to 1.5, then Eq. (3) and Eq. (4) can be rewritten as:

$$L_{i,0} = 1.5 d_{i,0} = 1.5 |Q_{i,1} P_i| = 1.5 |P_i Q_{i,3}| \tag{5}$$

$$|Q_{i,1} P_i| = |P_i Q_{i,3}| = d_{i,0} = \frac{2}{3} L_{i,0} \tag{6}$$

then the five control points of the cubic NURBS can be given as:

$$\begin{cases} Q_{i,0} = P_i - L_{i,0} \boldsymbol{a}_{i,0} \\ Q_{i,1} = P_i - \frac{2}{3} L_{i,0} \boldsymbol{a}_{i,0} \\ Q_{i,2} = P_i \\ Q_{i,3} = P_i + \frac{2}{3} L_{i,0} \boldsymbol{a}_{i,1} \\ Q_{i,4} = P_i + L_{i,0} \boldsymbol{a}_{i,1} \end{cases} \tag{7}$$

Where the $\boldsymbol{a}_{i,0}$ and $\boldsymbol{a}_{i,1}$ are the unit vectors of two adjacent segments.

## 2.2 Approximation error of transition curve

As illustrate in Fig. 1, the proposed cubic NURBS curve will cause the geometric error with respect to the original G01 linear segments which known as smoothing error. Since the inserted curve is symmetrical about the bisector corner, the maximal approximation error and curvature extreme all appear at the middle of the NURBS curve $C_i(u)$, so the maximal approximation error can be expressed as $\varepsilon = |P_i C_i(0.5)|$, calculated as Eq. (8) the derivation is given in Appendix A. And the curvature extreme $k_m$ can be analytically expressed as Eq. (9), the derivation is given in Appendix B.

$$\varepsilon = \frac{2 w_{i,1} L_{i,0} sin\left(\frac{\beta_i}{2}\right)}{3(w_{i,1} + w_{i,2})} \tag{8}$$

$$k_m = \frac{w_{i,2} sin(\beta_i)}{w_{i,1} L_{i,1} cos^3\left(\frac{\beta_i}{2}\right)} \tag{9}$$

Assume the maximum allowed geometric error of the CNC system is denoted as $\varepsilon_m$, thus the approximation error must satisfy $\varepsilon \le \varepsilon_m$, then the transition length $L_{i,0}$ is limited by:

$$L_{i,0} \le \frac{3(w_{i,1} + w_{i,2}) \varepsilon_m}{2 w_{i,1} sin\left(\frac{\beta_i}{2}\right)} \tag{10}$$

Meanwhile, the sum of the transition length of the two adjacent curves must not exceed the length of the shared linear segment length as expressed in Eq. (11):

$$\begin{cases} L_{i-1,1} + L_{i,0} \le |P_{i-1} P_i| \\ L_{i,1} + L_{i+1,0} \le |P_i P_{i+1}| \end{cases} \tag{11}$$

# 3 Transition length optimization

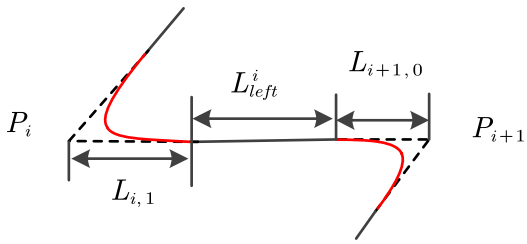


**Fig. 2** Length constrained corner smoothing

The maximal transition federate can be reached if the curvature extreme of every transition curve is the smallest. Once the transition curve type is defined, the curvature of the transition curve has to do with the constraint transition length and the corner angle of the adjacent linear segment. Normally, the corner angle of the adjacent linear segment will keep the same as the original NC, so the transition length $L_{i,0}$ is limited by Eq. (10).

$$\begin{cases} min \sum_{i=1}^{n} L^i_{left} \\ min \sum_{i=1}^{n} (L_{i+1,0} - L_{i,1}) \end{cases} \tag{12}$$

As illustrate in Fig. 2, In this literature, the optimal solution is to maximum use the linear segments of each corner or minimize the remained linear segments as expressed in Eq. (12), at the same time, the adjacent corner will not be polarized, that's to say, one corner consumed too much linear length, while the other only very short length can be used should not happen.

As can be seen from Eq. (10) ,the curvature extreme $k_m$ is inversely proportional to transition length. Unfortunately, this polarized case is inevitable in current PA strategy and asymmetrical methods. As illustrate in Fig. 3, according to Eq. (10) ,under the same approximation error, the smaller the corner angle of two adjacent linear segments requires the longer transition length, and generate the smaller curvature extreme, on contrary the larger corner angle cause larger curvature extreme, and in considering the real machining kinematic constraint ,the larger curvature extreme, requires the smaller feed speed. So, in processing short linear segment smoothing, one need to keep balance the transition length and curvature distribution, so that the feed speed sharp dropping can be mitigate in the sharp corner. In this paper to keep balance the transition length and curvature distribution is realized by minimizing the difference of the transition length of the adjacent two corners when eliminating the overlaps ,as expressed in Eq. (12).

## 3.1 The working of optimization method

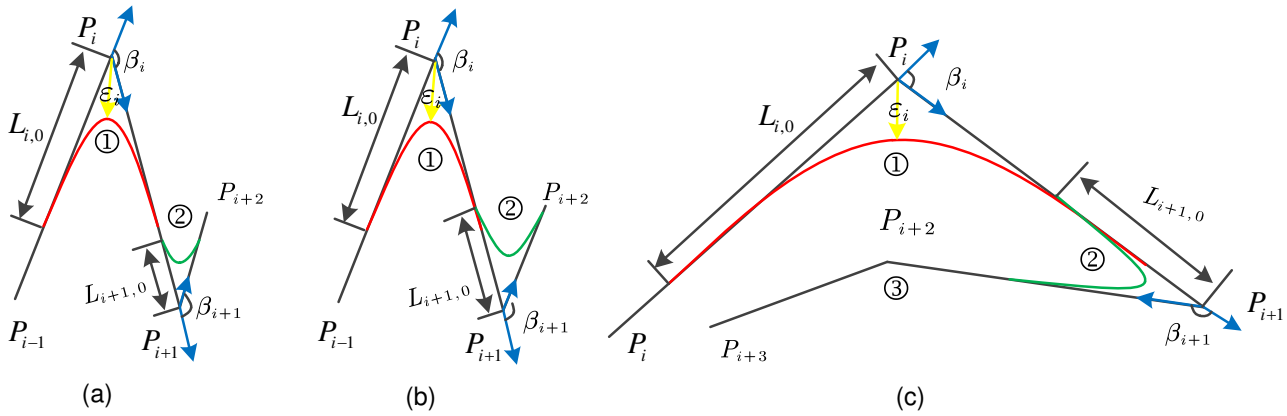


**Fig. 3** Transition length optimization

In the case of Fig. 3(a) the approximation error constraint transition lengths satisfy Eq. (13), then there no overlap happen in these adjacent two corners, so, in the proposed method ,the optimal transition length in corner ① will keep as the approximation error constraint transition length, instead of shrink it to $0.5|P_iP_{i+1}|$like FLCR.

$$\begin{cases} L_{i,0}+L_{i+1,0}<|P_iP_{i+1}| \\ L_{i,0}>0.5|P_iP_{i+1}| \end{cases} \tag{13}$$

In the case of Fig. 3(b), the approximation error constraint transition lengths satisfy Eq. (14), if just consider the approximation error constraint transition length, the overlap will happen here ,but in the proposed method ,after look ahead processing ,the pre-estimate transition length in corner ② may not over than $0.5|P_{i+1}P_{i+2}|$, thus, the optimal transition length of corner ① may keep as the approximation error constraint transition length, if need shrinking ,nevertheless ,the segment $P_iP_{i+1}$ is fully used in this case.

$$\begin{cases} L_{i,0}+L_{i+1,0}>|P_iP_{i+1}| \\ L_{i,0}>0.5|P_iP_{i+1}| \\ 0.5|P_iP_{i+1}|>|P_{i+1}P_{i+2}| \end{cases} \tag{14}$$

In the case of Fig. 3(c), the approximation error constraint transition lengths satisfy Eq. (15), here absolutely occurs overlap. After look ahead processing ,here assume the pre-estimate transition length corner ② is $L'_{i+1,0}$, then the optimal transition length $L'_{i,0}$ of corner ① is given by Eq. (16), thus mitigate curvature polarization of these two corners ,since corner ② is first considered, unlike PA strategy give too much portion of segment $P_iP_{i+1}$ to ① since the corner angle of corner ① is smaller than corner ② .

$$\begin{cases} L_{i,0}+L_{i+1,0}>|P_iP_{i+1}| \\ L_{i,0}>0.5|P_iP_{i+1}| \\ 0.5|P_iP_{i+1}|<|P_{i+1}P_{i+2}| \\ L_{i+1,0}>0.5|P_{i+1}P_{i+2}| \end{cases} \tag{15}$$

$$\begin{cases} L'_{i,0} = |P_iP_{i+1}| - L'_{i+1,0}, & L'_{i+1,0} < 0.5|P_iP_{i+1}| \\ L'_{i,0} = 0.5|P_iP_{i+1}|, & others \end{cases} \tag{16}$$

To real time the algorithm, and practical in long and short linear segments, the look-ahead segment number must be limited, there are two factors to stop the scanning process:

(1): the segment number reach the pre-defined maxim number.

(2): the following segments have no more influence on the previous transition curve, it happens when the two adjacent curves satisfy $L_{left}^{i} > 0$.

In this optimal model, once factor (1) is satisfied, then the last optimal transition length $l_{opt}$ was set to $mim\{0.5|P_nP_{n+1}|, L_{n,1}\}$, then backward to determine the previous optimal transition length. once the criteria (2) is satisfied, the last optimal transition length $l_{opt}$ was set to the approximation error constraint length $L_{i,1}$, then backward to determine the previous optimal transition length, thus can full use the linear segment, without linear parts remain. The transition length optimal algorithm is given in Appendix C.

# 4 Experimental works and result

## 4.1 Experimental features

**Table 1** Parameters used in path smoothing and characterizing the kinematics performance

| Parameter | Pentagram-shaped tool path | Corona tool path | Butterfly tool path |
|---|---|---|---|
| Approximation error | $0.8\mathrm{mm}$ | $20\mu\mathrm{m}$ | $10\mu\mathrm{m}$ |
| Chord error | $0.1\mu\mathrm{m}$ | $0.1\mu\mathrm{m}$ | $0.1\mu\mathrm{m}$ |
| Tangential acceleration limitation | $2000\mathrm{mm/s^2}$ | $1000\mathrm{mm/s^2}$ | $1000\mathrm{mm/s^2}$ |
| Normal acceleration limitation | $2000\mathrm{mm/s^2}$ | $1000\mathrm{mm/s^2}$ | $1000\mathrm{mm/s^2}$ |
| Normal jerk limitation | $80000\mathrm{mm/s^2}$ | $40000\mathrm{mm/s^2}$ | $80000\mathrm{mm/s^2}$ |
| Programmed feedrate | $5000\mathrm{mm/min}$ | $600\mathrm{mm/min}$ | $6000\mathrm{mm/min}$ |
| Interpretation period | $125\mu\mathrm{s}$ | $125\mu\mathrm{s}$ | $125\mu\mathrm{s}$ |
| Sampling period | $125\mu\mathrm{s}$ | $125\mu\mathrm{s}$ | $125\mu\mathrm{s}$ |

The proposed method has been coded in VS2010 using C++ programming and implemented on a PC with i7-8550U CPU and 8 G RAM. Both the computer simulation and the experimental machining testing were carried out for testing and validation purpose. In our experiments, corner rounding for the typical Pentagram-shaped tool path and the coronal tool path are tested in simulation, and corner rounding of bat-shaped tool path is tested by real machining experiment. In these tests, the proposed method is compared with the classical FLCR method [5] and PA strategy [23] for rounding corners. The Pentagram-shaped profile as is shown in Fig. 4(a) consists of 10 corners, 5 obtuse corners and 5 acute corners, and all the linear segments with the same length. The machining starts from the top corner and in clockwise direction, so only 4 acute corners and 5 obtuse corners are rounded in this experiment. The coronal tool path is shown in Fig. 7(a) with 160 corners and typical short linear segments, starts from the bottom right of the top coronal and in counterclockwise direction, and because the NC tool point only keeps three decimal places, the coronals are not totally the same. The real machining butterfly tool path has 198 corners as show in figure Fig. 12(a), the machining test start

from the most top point of the shape. To ensure the tested methods share the kinematically similar performances, a complete S-shaped federate scheduling method [30] is used to schedule the feedrate profile of the smoothed tool path. The parameters used in characterizing the kinematics performance of the tool paths for testing are listed in Table 1.

## 4.2 Comparison with the FLCR method

In this experiment, at the acute corners, the proposed and the FLCR obtain the same curvature extreme as can be seen in Fig. 5, it is because the approximation constraint transition length in acute corners are less than half length of the linear segment, so while implementing segment length constraint processing, these two methods made no adjustment to the approximation constraint transition length. For the FLCR method, the approximation constraint transition length not exceed half length of the adjacent two linear segment, so no need to adjust it, for the proposed method, although the sum of the approximation constraint transition length of the adjacent acute corner and obtuse corner exceed the shared linear segment, but because the approximation constraint transition length not exceed half length of the linear segment, so the optimal transition length of the acute corners keeps the same as the approximation error constraint transition length.

However, there are great difference between these two methods while rounding the obtuse corners, which can be seen in the magnified view of the NC segment and rounded tool orbit in Fig. 4(c), and the curvature extremes of transition curves at each corner are showed in Fig. 5. For the FLCR method, the transition lengths cannot overpass half of the linear segments, even when there will be linear segment left, while the proposed method can fully use the linear segments to achieve lower curvature at the obtuse corners. By analysis, the proposed method achieved a significant curvature decrease of 17.64%. From the view of feedrate profile in Fig. 6, it is very clear that the decrease of curvature results in the increase of federate substantially. This makes the machining time decreased from 998.875ms to 979.25ms, without doubt, the proposed optimal method improved the machining efficiency further. In the aspect of algorithm complexity, one can see from the given pseudo code in Appendix C, there is no complex mathematical operation or recursively calculation compared to FLCR.

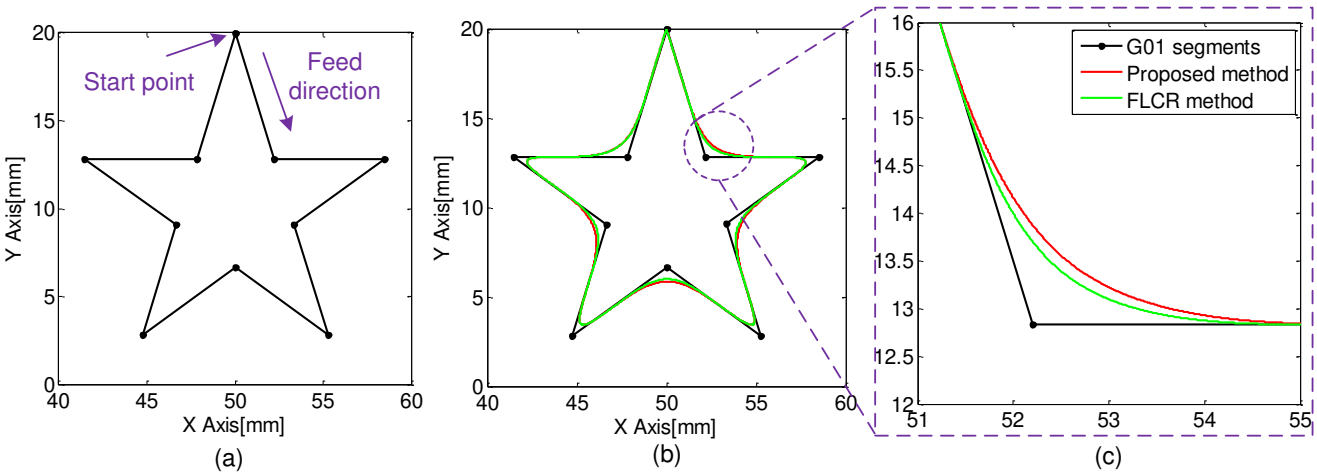


**Fig. 4** Smoothing the Pentagram-shaped tool path. (a) Pentagram-shaped linear segments for testing; (b) NC segments and smoothed tool paths by proposed method and FLCR method; (c) magnified view of smoothed tool path

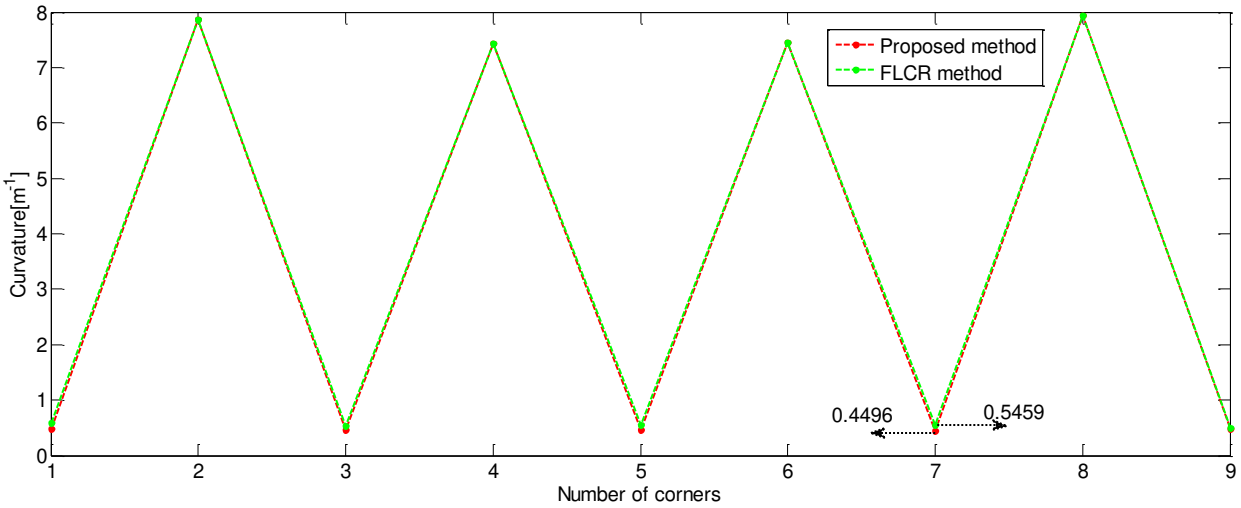


**Fig. 5** Curvature extreme of transition curves of the proposed method and FLCR method in Pentagram-shaped tool path smoothing

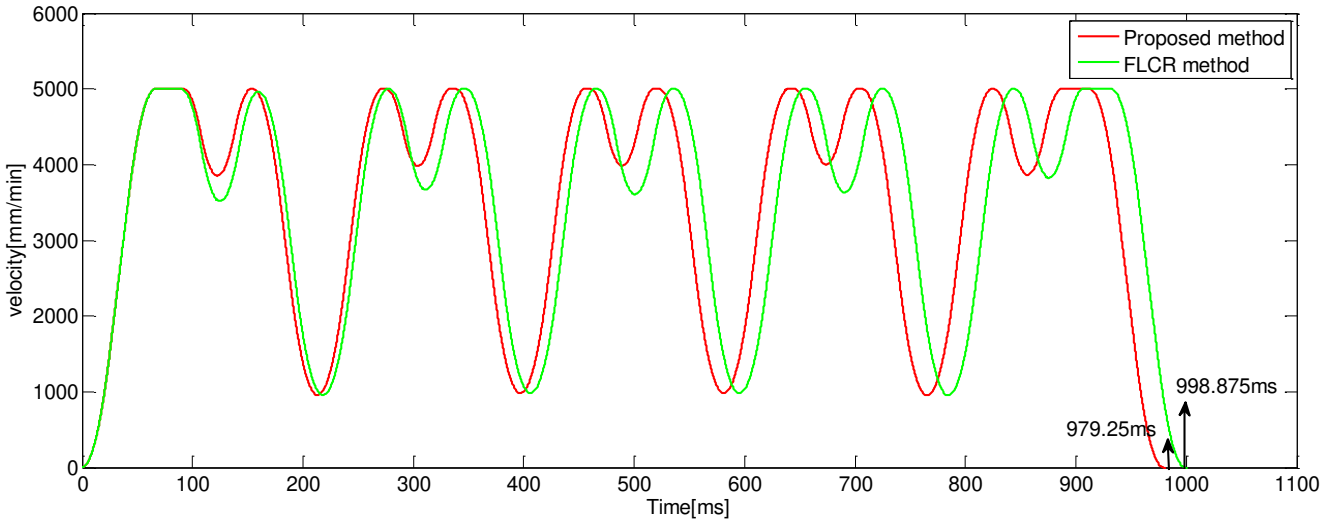


**Fig. 6** Feedrate profiles of proposed method and FLCR method in Pentagram-shaped tool path smoothing

## 4.3 Comparison with the PA strategy

In this simulation comparison ,a coronal tool path is used to compare the proposed method with the PA strategy in [23]. With the predefined approximation error tolerance as show in Table 1 ,the transition curve overlap happened in every coronal. Thus, the overlap eliminating process between the proposed method and PA strategy is compared in detail to demonstrate the advantage of the proposed method in this paper. The smoothed tool path by proposed method and PA strategy are pictured in Fig. 7(b). Fig. 7(c) shows the partial magnified view of tool path. To compare the control points distribution of PA strategy and the proposed method, two partial enlarged view of control points are pictured in Fig. 8 and Fig. 9. The smoothed curvature extreme of each corner is pictured in Fig. 10, it is very clear that the proposed method in some corners have lager extreme than PA strategy, while in some corners smaller than PA strategy ,but overall ,the proposed curvature extreme is smaller than PA strategy, and the proposed method have more gentle curvature distribution as can be seen in the magnified view in Fig. 10. And from the perspective of machining speed profile in Fig. 11, the largest and smallest speed of proposed method are higher than PA strategy, thus the proposed method has higher efficiency than PA strategy.

By comparison the control points distribution of the proposed method and PA method, which are pictured in Fig. 8 and Fig. 9 respectively, it can be seen that by proposed method there are only in segment $P_{11}P_{12}$ and $P_{15}P_{16}$ remained linear portion after smoothing, but by PA strategy the linear portion appeared not only in $P_{11}P_{12}$ and $P_{15}P_{16}$, but also in $P_8P_9$ and $P_9P_{10}$. Since the machining direction is counterclockwise, by PA strategy the sum of approximation error constraint length of corner $P_8$ and $P_9$ is exceed the length of segment $P_8P_9$, so PA strategy will shrink the transition length of corner $P_8$ and $P_9$ proportionally to eliminate the potential overlap, when smoothing corner $P_9$, there is overlap in segment $P_9P_{10}$, so the transition length of corner $P_9$ and $P_{10}$ adjusted proportionally to eliminate the overlap ,and corner $P_9$ get a portion from segment $P_9P_{10}$ which is shorter than the linear segment left in segment $P_8P_9$ after smoothing corner $P_8$, thus found there linear part remained in segment $P_8P_9$. In our proposed method ,with the pre-read processing ,the transition length of corner $P_9$ was pre-constrained by the following corners $P_{10}$, $P_{11}$ etc., and while analyzing the potential overlap of corner $P_8$ and $P_9$, the pre-constrained transition length of corner $P_9$ is obviously less than half of the segment $P_8P_9$, because $|P_9P_{10}|<|P_8P_9|$, so here the pre-constrained transition length of corner $P_9$ is kept ,the transition length of corner $P_8$ is adjusted to eliminate the overlap, and after the smoothing of corner $P_8$, the remained linear part of segment $P_8P_9$ will be fully used in corner $P_9$ smoothing, thus no linear portion left in segment $P_8P_9$, and the proposed method get smaller curvature extreme in corner $P_8$ compared to PA strategy. In segment $P_9P_{10}$ also has linear portion left by PA strategy, it is the same situation as smoothing corner $P_8$ and $P_9$, and linear portion left in segment $P_8P_9$. In corner $P_{10}$ our method gets smaller curvature but in corner $P_{11}$ gets lager curvature compared to PA strategy, it is because the approximation error constraint transition length in corner $P_{10}$ and $P_{11}$ are exceed $0.5|P_{10}P_{11}|$, and $|P_{10}P_{11}|<|P_9P_{10}|$ ,and $|P_{10}P_{11}|<|P_{11}P_{12}|$, so corner $P_{11}$ get $0.5|P_{10}P_{11}|$ as transition length ,so also the corner $P_{10}$, but by PA strategy ,while eliminating the overlap in segment $|P_{10}P_{11}|$, corner $P_{11}$ get a portion more than half of the segment $P_{10}P_{11}$, let it be noted as $L_{11,0}>0.5|P_{10}P_{11}|$ , and corner $P_{11}$ and $P_{12}$ have the same corner angle of the two adjacent linear segments, so by PA strategy corner $P_{11}$ can get half of the segment $P_{11}P_{12}$, let it be noted as $L_{11,1}=0.5|P_{11}P_{12}|$, but $L_{11,0}<L_{11,1}$, so the segment length constraint transition length of $P_{11}$ is $L_{11,0}$, thus PA strategy get smaller curvature in corner $P_{11}$ and $P_{12}$ ,but larger curvature in $P_{10}$ compared to our method. In segment $P_{11}P_{12}$ there is linear portion left is because $|P_{10}P_{11}|<|P_{11}P_{12}|$, and the remained linear segment length of $P_{10}P_{11}$ after smoothing corner $P_{10}$ is less than $0.5|P_{11}P_{12}|$, and when smoothing corner $P_{12}$ the transition length also limited by the segment length of $P_{12}P_{13}$, corner $P_{12}$ get less than half of segment $P_{11}P_{12}$ as transition length ,thus has linear portion left in segment $P_{11}P_{12}$. In segment $P_{15}P_{16}$ also has linear portion left after smoothing, it is the same situation as segment $P_{11}P_{12}$.

By analysing magnified curvature extreme distribution in Fig. 10 and speed profile in Fig. 11, the bottom speed happens in $P_{10}$ and $P_{13}$, the top speed appears in segment $P_{15}P_{16}$ (also can be said before $P_8$), it is clear our method has higher speed in these point ,because our method get smaller curvature extreme in these corners. In corners $P_{11}$ and $P_{12}$, and $P_{14}$ although PA strategy has smaller curvature extreme, but the linear segments in these related segments are very short, there are not enough distance to reach the curvature extreme allowed maxim speed under certain speed acceleration, that's why the whole machining time is longer than our method. The proposed optimal method not only fully use the linear segments but also keep the curvature extreme distribution gentler, makes the machining time decrease from 2.477 s to 2.367 s, thus the machining efficiency is improved 4.44% compared to PA strategy.

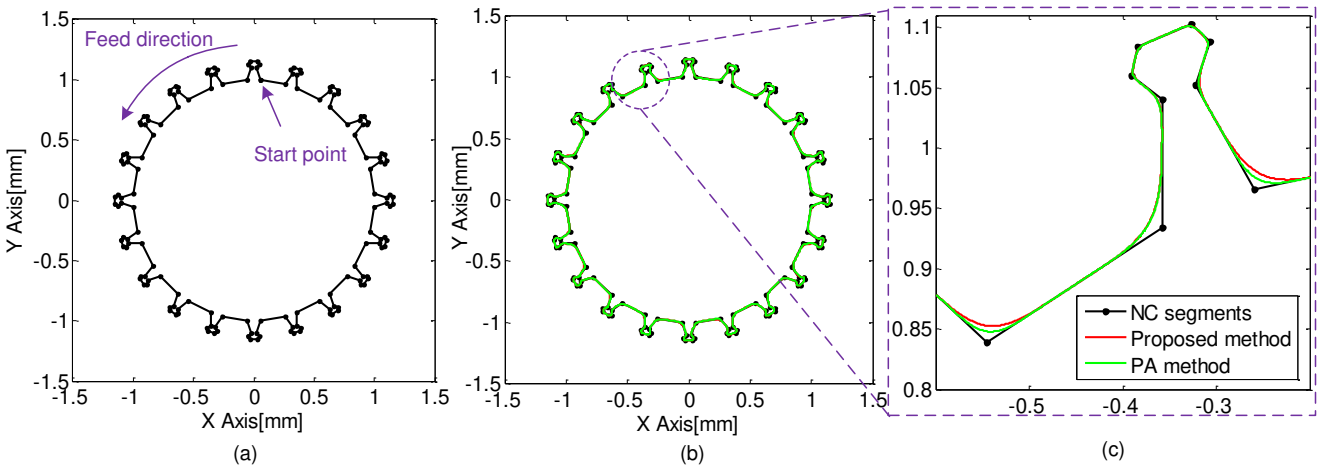


**Fig. 7** Smoothing of coronal tool path. (a) coronal linear segments; (b) NC segments and smoothed tool paths by proposed method and PA strategy; (c) magnified view of smoothed tool path

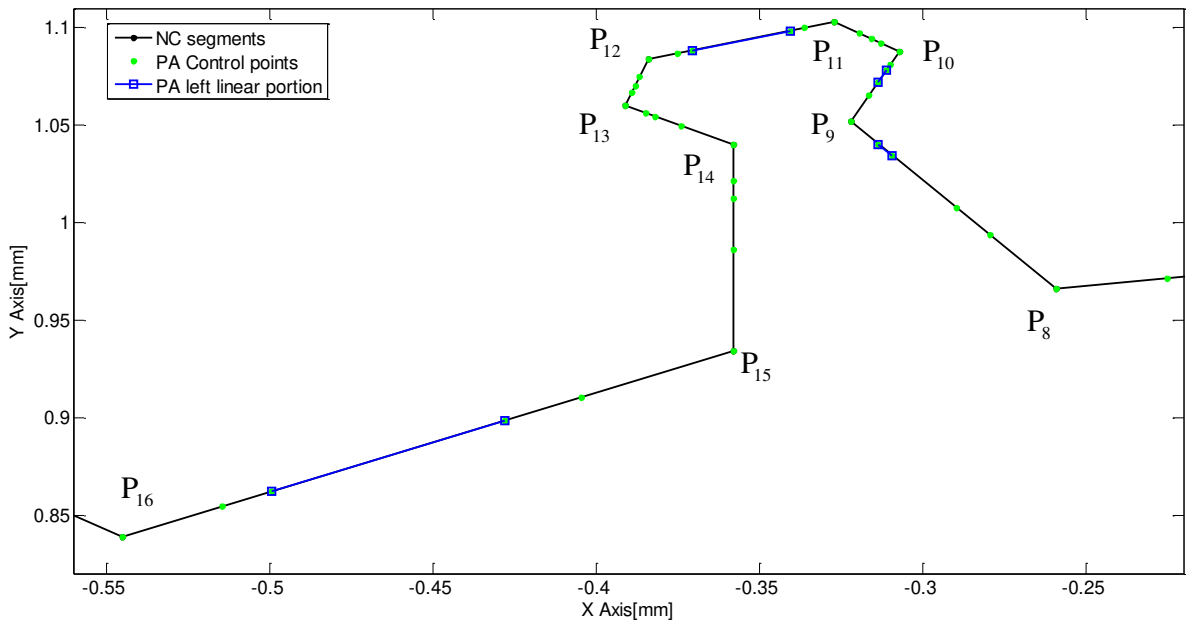


**Fig. 8** Partial magnified view of control points distribution of PA strategy in coronal tool path smoothing

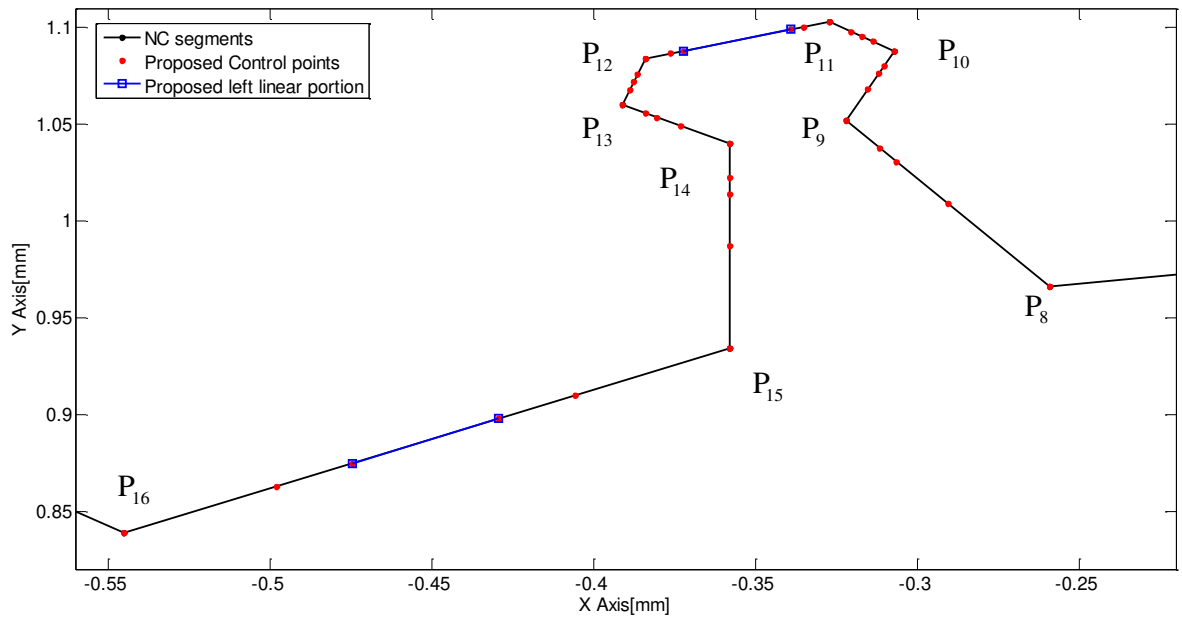


**Fig. 9** Partial magnified view of control points distribution of proposed method in coronal tool path smoothing

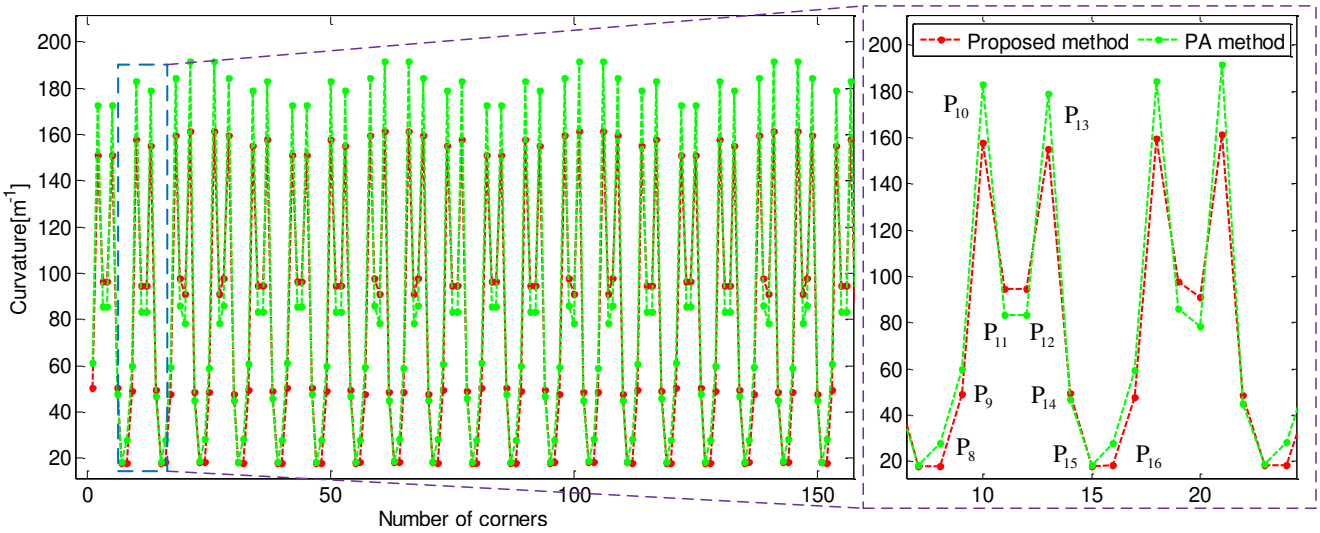


**Fig. 10** Curvature extreme of transition curves of the proposed method and PA strategy in coronal tool path smoothing

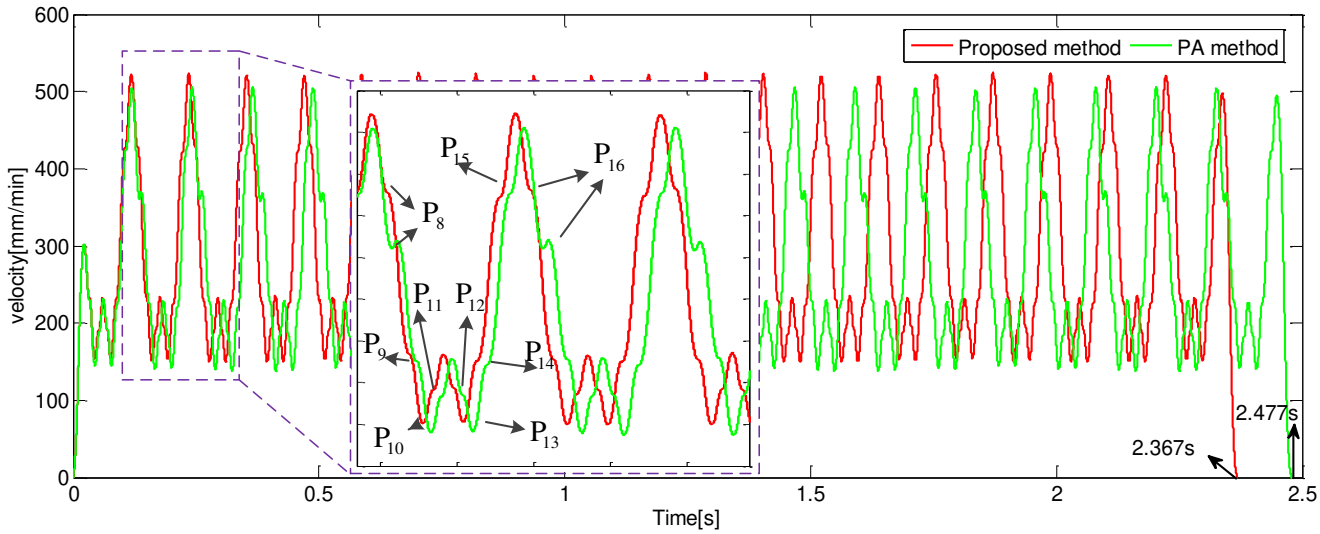


**Fig. 11** Feedrate profiles of proposed method and PA strategy in coronal tool path smoothing

## 4.4 Real Machining experiment

In this real machining test, a butterfly tool path is performed on a CNC machine to demonstrate the advantage of our method in short linear segments smoothing and the real-time performance, compared with the PA strategy, but different from the recursive function in [22, 23], the proportional shrink factor is obtained mathematically, in Appendix D. thus, the PA strategy can be implicated in the real-time machining test. With the specified approximation error $\varepsilon_0 = 10\mu m$, the transition overlaps occurred in some places where the tool points are dense. In this test, all the transition overlaps are eliminated by the proposed optimal mothed and PA strategy and the smoothed tool orbits are shown in Fig. 12(a). The machining results show in Fig. 12(b). From the magnified view of the smoothed tool orbits in Fig. 12(c), it is very clear that in the sharp corner the proposed method have smaller curvature than PA strategy ,while the adjacent corner which before the sharp one ,PA strategy get smaller curvature, this is because here the transition overlap happened and the proposed method balanced the distribution of the length of the linear segments.

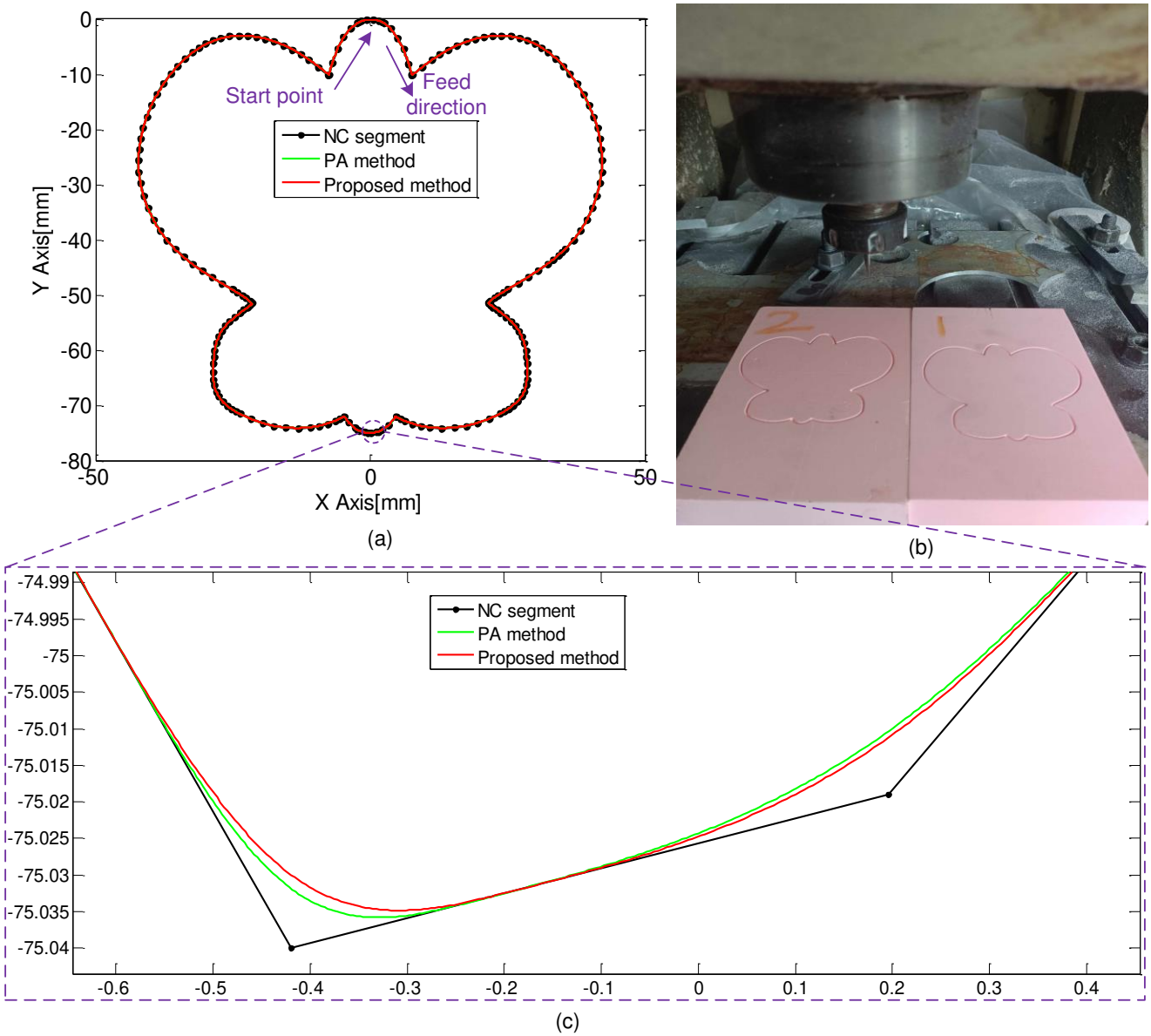


**Fig. 12** Smoothing of butterfly tool path. (a) butterfly linear segments and smoothed tool paths by proposed method and PA strategy; (b) the machining results; (c) magnified view of smoothed tool path

As shown in Fig. 13 the machining velocity profile of both PA strategy and the proposed method are pictured respectively. From the magnified view of the velocity profile of the red circle circled portion in Fig. 13(a), there is fluctuation in the place where the interpolation velocity should be kept in $6000mm/min$, it is because the pulse rounding accuracy, so, 1 pulse fluctuation in the moving axis is inevitable ,and in this test 1mm distance equals 100000 interpolation pulses, so the maxim rounding fluctuation could be $\sqrt{2}$ pulse for two axis-linkage, thus the fluctuation could be $6.78823mm/min$ in one sample period, as calculated in Eq. (17). And the feedback velocity profile is always delayed to the interpolation velocity, and has fluctuation, and the amplitude is larger than the interpolation velocity profile, it is normal, it also has to do with the parameter of the servo and the performance of servo system, here will not discuss about it, for it's out of the scope of the paper.

$$60 * \left(\sqrt{2}/100000\right) * 8000 = 6.78823 \tag{17}$$

From the magnified view of velocity profile in Fig. 13(b), in the none-overlap corner, the PA strategy and the proposed method have the same velocity level, that means the curvature is the same, while in the overlap segments, the proposed

method can obtain balanced transition length thus the curvature extreme won't be polarized ,that means the proposed method have higher curvature extreme level in the curvature bottom ,but lower curvature extreme level in the curvature top compared to PA strategy. Thus, the proposed method obtained higher machining efficiency than PA strategy, which decreased machining time from 4.9785s to 4.8985s.

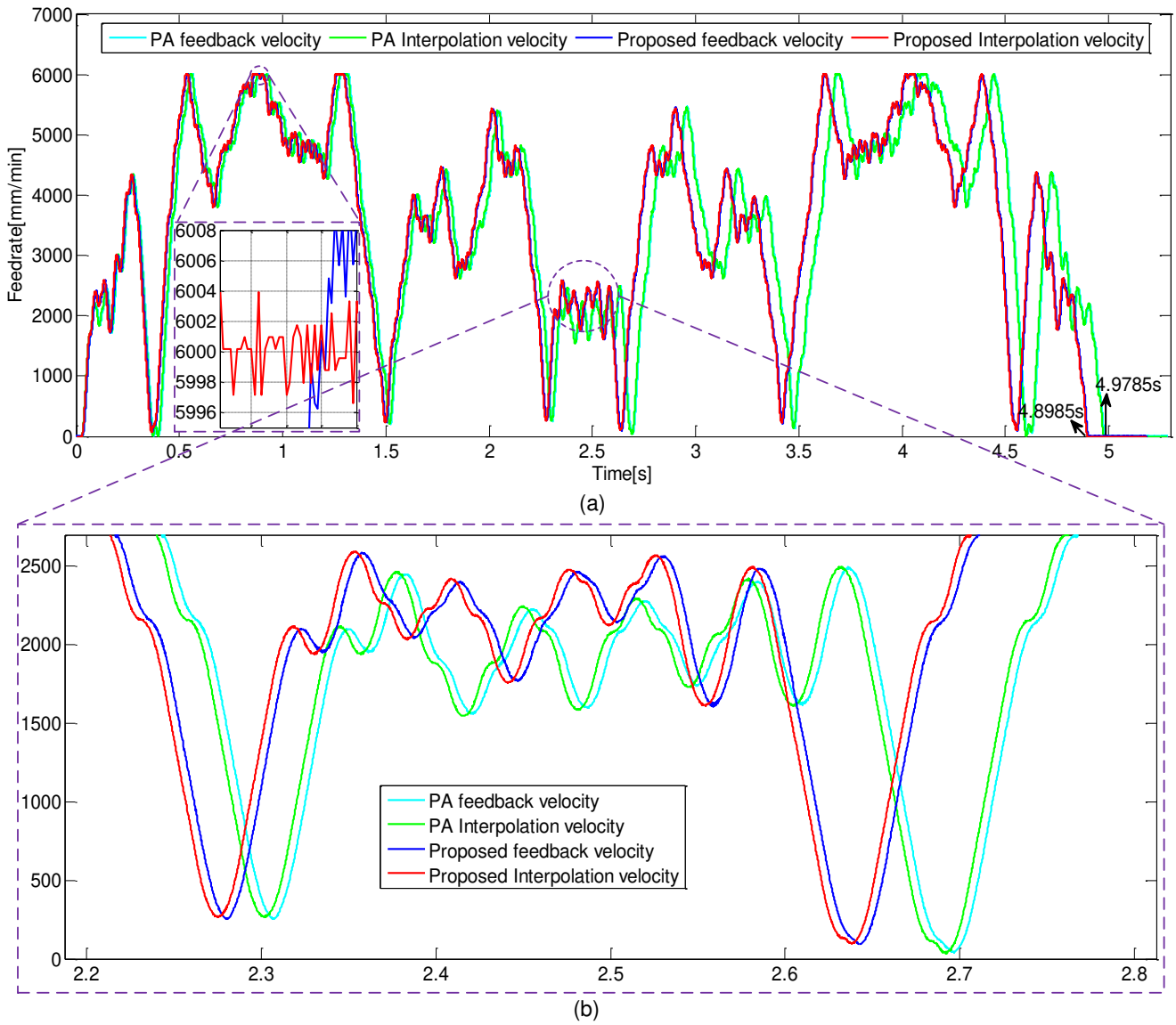


**Fig. 13** Velocity profile in butterfly tool path smoothing. (a) Interpolation and feedback velocity profile; (b) magnified view of velocity profile

# 5 Conclusion

In this paper, a local corner rounding algorithm with look ahead optimal strategy for short and long G01 linear segments is presented. The control points of cubic NURBS transition curve are parameterized to make the curve symmetrical about the included angle bisector, and $G^2$ continuity of the smoothed tool path is guaranteed along the entire tool path and within specified geometric error. Based on the characteristic of symmetrical curve, the curvature extreme is mathematically presented, which can be used in machining velocity planning to obtain the kinematic constraint speed. The transition length optimal algorithm in this paper not only eliminates the transition overlap, but also can fully use the linear segment and avoid polarization or unbalanced ratio of curvature extreme of the adjacent curves while eliminating the overlap, thus decreasing the curvature extreme as much as possible, at the same time get more smoothed

curvature extreme distribution especially in short linear segments NC machining. In comparison with traditional FLCR our proposed method achieved a significant curvature decrease of 17.64%. In comparison with PA strategy the proposed method obtained balanced curvature distribution and an efficiency improvement of 4.44%. In addition, our method can work well in 125us control period real-time CNC system which was verified in real CNC machining experiment.

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**Authors' Contributions** Yusoff Nukman, Hwa Jen Yap did supervision. Chunlin Lei contributed to the study conception, code the algorithm and perform the simulation experiment, real machining experiments were performed by Mingbing Luo. Chunlin Lei, Yusoff Nukman and Mingbing Luo discussed the experiments result. Yusoff Nukman, Hwa Jen Yap and Ali H. Abdelrazek commented on previous versions of the manuscript, all authors read and approved the final manuscript.

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**Data and Material availability** Not applicable.

**Code availability** Not applicable.

# Declarations

**Ethics approval** This manuscript is the authors' own original work, which has not been previously published elsewhere.

**Consent to participate** Not applicable.

**Consent for publication** Not applicable.

**Conflict of interest** The authors declare no competing interests.

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**Appendix A.** The calculation of max approximation error of NURBS smoothing curve

At first according to Eq.(2) the basis function $N_{i,p}(u)$ can be calculated as following:

$$N_{0,0}=N_{1,0}=N_{2,0}=0,$$

$$N_{3,0}=\begin{cases} 1 & 0\le u<0.5 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{4,0}=\begin{cases} 1 & 0.5\le u<1 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{5,0}=N_{6,0}=N_{7,0}=0, \quad -\infty<u<\infty$$

$$N_{1,1}=\frac{u-0}{0-0}N_{1,0}+\frac{0-u}{0-0}N_{2,0}=0, \quad -\infty<u<\infty$$

$$N_{2,1}=\frac{u-0}{0-0}N_{2,0}+\frac{0.5-u}{0.5-0}N_{3,0}=\begin{cases} 2(0.5-u) & 0\le u<0.5 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{3,1}=\frac{u-0}{0.5-0}N_{3,0}+\frac{1-u}{1-0.5}N_{4,0}=\begin{cases} 2u & 0\le u<0.5 \\ 2(1-u) & 0.5\le u<1 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{4,1}=\frac{u-0.5}{1-0.5}N_{4,0}+\frac{1-u}{1-1}N_{5,0}=\begin{cases} 2(u-0.5) & 0.5\le u<1 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{5,1}=\frac{u-1}{1-1}N_{5,0}+\frac{1-u}{1-1}N_{6,0}=0, \quad -\infty<u<\infty$$

$$N_{6,1}=\frac{u-1}{1-1}N_{6,0}+\frac{1-u}{1-1}N_{7,0}=0, \quad -\infty<u<\infty$$

$$N_{0,2}=\frac{u-0}{0-0}N_{0,1}+\frac{0-u}{0-0}N_{1,1}=0, \quad -\infty<u<\infty$$

$$N_{1,2}=\frac{u-0}{0-0}N_{1,1}+\frac{0.5-u}{0.5-0}N_{2,1}=\begin{cases} 4(0.5-u)^2 & 0\le u<0.5 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{2,2}=\frac{u-0}{0.5-0}N_{2,1}+\frac{1-u}{1-0}N_{3,1}=\begin{cases} 4u(0.5-u)+2u(1-u) & 0\le u<0.5 \\ 2(1-u)^2 & 0.5\le u<1 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{3,2}=\frac{u-0}{1-0}N_{3,1}+\frac{1-u}{1-0.5}N_{4,1}=\begin{cases} 2u^2 & 0\le u<0.5 \\ 2u(1-u)+4(1-u)(u-0.5) & 0.5\le u<1 \\ 0 & \text{ohterwise} \end{cases}$$

$$N_{4,2}=\frac{u-0.5}{1-0.5}N_{4,1}+\frac{1-u}{1-0.5}N_{5,1}=\begin{cases}4(u-0.5)^2 & 0.5\le u<1\\ 0 & \text{ohterwise}\end{cases}$$

$$N_{5,2}=\frac{u-1}{1-1}N_{5,1}+\frac{1-u}{1-1}N_{6,1}=0\,,\quad -\infty<u<\infty$$

$$N_{0,3}=\frac{u-0}{0-0}N_{0,2}+\frac{0.5-u}{0.5-0}N_{1,2}=\begin{cases}8(0.5-u)^3 & 0\le u<0.5\\ 0 & \text{ohterwise}\end{cases}$$

$$N_{1,3}=\frac{u-0}{0.5-0}N_{1,2}+\frac{1-u}{1-0}N_{2,2}=\begin{cases}8u(0.5-u)^2+2u(1-u)(2-3u) & 0\le u<0.5\\ 2(1-u)^3 & 0.5\le u<1\\ 0 & \text{ohterwise}\end{cases}$$

$$N_{2,3}=\frac{u-0}{1-0}N_{2,2}+\frac{1-u}{1-0}N_{3,2}=\begin{cases}4u^2(0.5-u)+4u^2(1-u) & 0\le u<0.5\\ 4u(1-u)^2+4(1-u)^2(u-0.5) & 0.5\le u<1\\ 0 & \text{ohterwise}\end{cases}$$

$$N_{3,3}=\frac{u-0}{1-0}N_{3,2}+\frac{1-u}{1-0.5}N_{4,2}=\begin{cases}2u^3 & 0\le u<0.5\\ 2u^2(1-u)+(1-u)(u-0.5)(12u-4) & 0.5\le u<1\\ 0 & \text{ohterwise}\end{cases}$$

$$N_{4,3}=\frac{u-0.5}{1-0.5}N_{4,2}+\frac{1-u}{1-1}N_{5,2}=\begin{cases}8(u-0.5)^3 & 0.5\le u<1\\ 0 & \text{ohterwise}\end{cases}$$

then according to Eq. (1) and Eq. (7) can get $C_i(0.5)$ as :

$$C_i(0.5)=\frac{\sum_{j=0}^{4}N_{j,3}(0.5)w_{i,j}Q_{i,j}}{\sum_{j=0}^{4}N_{j,3}(0.5)w_{i,j}}=\frac{0.5w_{i,1}(Q_{i,1}+Q_{i,3})+w_{i,2}Q_{i,2}}{w_{i,1}+w_{i,2}}$$

$$=\frac{0.5w_{i,1}(P_i-\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,0}+P_i+\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,1})+w_{i,2}P_i}{w_{i,1}+w_{i,2}}=P_i+\frac{w_{i,1}L_{i,0}(\boldsymbol{a}_{i,1}-\boldsymbol{a}_{i,0})}{3(w_{i,1}+w_{i,2})}$$

Thus

$$\varepsilon=P_iC_i(0.5)=\left|\frac{w_{i,1}L_{i,0}(\boldsymbol{a}_{i,1}-\boldsymbol{a}_{i,0})}{3(w_{i,1}+w_{i,2})}\right|=\frac{2w_{i,1}L_{i,0}sin\,(\frac{\beta_i}{2})}{3(w_{i,1}+w_{i,2})}$$

**Appendix B.** The calculation of max curvature extreme of NURBS curve

Let $\mathrm{A}(u)=\sum_{j=0}^{4}N_{j,3}(u)w_{i,j}Q_{i,j}, W(u)=\sum_{j=0}^{4}N_{j,3}(u)w_{i,j}$, then

$$\mathrm{C}(u)=\frac{A(u)}{W(u)}$$

$$C'(u)=\frac{W(u)A'(u)-W'(u)A(u)}{W(u)^2}=\frac{W(u)A'(u)-W'(u)W(u)C(u)}{W(u)^2}=\frac{A'(u)-W'(u)C(u)}{W(u)}.$$

$$C''(u)=\frac{A''(u)-W''(u)C(u)-W'(u)C'(u)}{W(u)}$$

according to the derivation equation of Appendix A

$$A_i(u)|_{u=0.5}=0.25w_{i,1}(Q_{i,1}+Q_{i,3})+0.5w_{i,2}Q_{i,2}$$

$$W_i(u)|_{u=0.5}=0.25(w_{i,1}+w_{i,3})+0.5w_{i,2}=0.5(w_{i,1}+w_{i,2})$$

$$C_i(u)|_{u=0.5}=\frac{0.5w_{i,1}(Q_{i,1}+Q_{i,3})+w_{i,2}Q_{i,2}}{w_{i,1}+w_{i,2}}$$

$$A'(u)|_{u=0.5}=(-6(1-u)^2w_{i,1}Q_{i,1}+(24u^2-36u+12)w_{i,2}Q_{i,2}+(-42u^2+48u-12)w_{i,3}Q_{i,3})|_{u=0.5}$$

$$=-1.5w_{i,1}Q_{i,1}+1.5w_{i,3}Q_{i,3}$$

$$W'(u)|_{u=0.5}=0$$

$$A''(u)|_{u=0.5}=(12(1-u)w_{i,1}Q_{i,1}+(48u-36)w_{i,2}Q_{i,2}+(-84u+48)w_{i,3}Q_{i,3})|_{u=0.5}$$

$$=6w_{i,1}(Q_{i,1}+Q_{i,3})-12w_{i,2}Q_{i,2}$$

$$W''(u)|_{u=0.5}=6w_{i,1}+6w_{i,3}-12w_{i,2}$$

$$C'(u)|_{u=0.5}=\left.\frac{A'(u)-W'(u)C(u)}{W(u)}\right|_{u=0.5}=\frac{-1.5w_{i,1}Q_{i,1}+1.5w_{i,3}Q_{i,3}}{0.5(w_{i,1}+w_{i,2})}$$

$$=\frac{1.5w_{i,1}(Q_{i,3}-Q_{i,1})}{0.5(w_{i,1}+w_{i,2})}=\frac{1.5w_1(P_i+\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,1}-P_i+\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,0})}{0.5(w_{i,1}+w_{i,2})}$$

$$=\frac{w_{i,1}L_{i,0}}{0.5(w_{i,1}+w_{i,2})}(\boldsymbol{a}_{i,1}+\boldsymbol{a}_{i,0})$$

$$C''(u)|_{u=0.5}=\frac{A''(u)-W''(u)C(u)}{W(u)}$$

$$=\frac{6w_{i,1}(Q_{i,1}+Q_{i,3})-12w_{i,2}Q_{i,2}-12(w_{i,1}-w_{i,2})\frac{0.5w_{i,1}(Q_{i,1}+Q_{i,3})+w_{i,2}Q_{i,2}}{w_{i,1}+w_{i,2}}}{0.5(w_{i,1}+w_{i,2})}$$

$$=\frac{6w_{i,1}(P_i-\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,0}+P_i+\frac{2}{3}L_{i,0}\boldsymbol{a}_{i,1})-12w_{i,2}P_i-12(w_{i,1}-w_{i,2})[P_i+\frac{w_{i,1}L_{i,0}(\boldsymbol{a}_{i,1}-\boldsymbol{a}_{i,0})}{3(w_{i,1}+w_{i,2})}]}{0.5(w_{i,1}+w_{i,2})}$$

$$=\frac{16w_{i,1}w_{i,2}L_{i,0}}{(w_{i,1}+w_{i,2})^2}(\boldsymbol{a}_{i,1}-\boldsymbol{a}_{i,0})$$

So, the curvature extreme can be expressed as:

$$k=\frac{|C'(0.5)\times C''(0.5)|}{|C'(0.5)|^3}=\frac{\frac{w_{i,1}L_{i,0}}{0.5(w_{i,1}+w_{i,2})}\frac{16w_{i,1}w_{i,2}L_{i,0}}{(w_{i,1}+w_{i,2})^2}|(\boldsymbol{a}_{i,1}+\boldsymbol{a}_{i,0})\times(\boldsymbol{a}_{i,1}-\boldsymbol{a}_{i,0})|}{(\frac{4w_{i,1}L_{i,0}cos(\frac{\beta_i}{2})}{w_{i,1}+w_{i,2}})^3}$$

$$=\frac{2w_{i,2}sin(\beta_i)}{w_{i,1}L_{i,0}cos^3(\frac{\beta_i}{2})}$$

**Appendix C.** Transition length optimal algorithm

Input: Predetermined max approximation error $\varepsilon_m$ specifying the designed accuracy of the algorithm; specify the max pre-read segment number N (to real-time the algorithm, the look ahead number should be limited); data points $\{P_i\in R^3, 0\le i\le N\}$ of linear tool path.

**Output:** approximation error and linear segment length constraint NURBS curve transition length $l_{opt}$ and remained linear length $L_{left}$ after transition.

Step 0:

Calculate the approximation error constrained transition length $l_i$ of each corner which satisfies Eq.(10).

**Step 1:**

(1) Set optimal transition length $l_{opt}=0$, $K=0$.

(2) For $i=1:1:N-2$

(3) If $l_i+l_{i+1}>|P_iP_{i+1}|$

(4) Continue

(5) Else

(6) Break

(7) Endif

(8) End for

(9) $K=i$

(10) $l_{opt}=l_K$

**Step 2:**

(1) For $i=K:-1:2$

(2) If $l_{opt}+l_{i-1}>|P_{i-1}P_i|$

(3) If $l_{opt}>0.5|P_{i-1}P_i|$ && $l_{i-1}>0.5|P_{i-1}P_i|$

(4) $l_{opt}=0.5|P_{i-1}P_i|$

(5) Else if $l_{opt}\ge 0.5|P_{i-1}P_i|$

(6) $l_{opt}=l_{i-1}$

(7) Else

(8) $l_{opt}=|P_{i-1}P_i|-l_{opt}$

(9) Endif

(11) Endif

(12) Else

(13) $l_{opt} = l_{i-1}$

(14) Endif

(15) End for

(16) $L_{left} = |P_0 P_1| - l_{opt}$

**Appendix D.** Mathematical expression of shrinking factor

In [23] the error constraint for transition curve is denoted $h_i$ ,the segments length is denoted $L_i$ ,the overlap criteria is $\gamma_{i-1} h_{i-1} + \gamma_i h_i \leq L_i$, when the criteria is not satisfied then set $h_{i-1} = \lambda h_{i-1}$ && $h_i = \lambda h_i$ iteratively until the criteria is satisfied ,where $0 < \lambda < 1$ . Here assume after n times iteration, the criteria is satisfied, then: $(\gamma_{i-1} h_{i-1} + \gamma_i h_i)\lambda^n \leq L_i$.

Let $\rho = \dfrac{L_i}{\gamma_{i-1} h_{i-1} + \gamma_i h_i}$ , thus the proportional shrink factor is obtained mathematically, it is even more accurate than the iteration algorithm in [23].